

Introduction to Calculations of Heat loss on piping

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1 Introduction

At some point during one of the project I was a part of, it was necessary to check if the chosen thickness of insulation for long CIP water loops was enough, to keep the temperature control and over certain threshold. It turned out that one of calculations were done in a rush, resulting in wrong decision of making insulation thicker. Just in time, one of senior engineers stepped in and stopped the company from making somewhat pricy mistake.

This senior engineer, later send us the calculaitons to check it, write down things that were wrong and report back to him, to have a conversation about our findings. This way we could learn to spot errors. Because I like calculations and love learning about things inherently related to mathematics it was a match made in heaven. At this point in time I have not started my introductory book to mechanics of heat transfer by great J. Cengel, so it wasn't something I could jump right into. Thus, I took my time, and in around 2 weeks felt somewhat comfortable, at least with concepts (side note: I was especially taken by transient analysis and eigendecomposition of differential equations in plug flow problems, amazing, it might be something to write about) in steady state analysis to begin deriving equations and testing my knowledge.

The original calculations were made using iterative method, which is fine if the iterations are "fairly" small or the change in the underlying function is relatively insignificant or, even better, constant.

2 Notation

The functions we will be dealing with depends inherently on a lot of variables, some of them are constant no matter the distance the fluid traveled, others are not. In general the notation for each function F can be presented as:

$$F(x, \mathbf{v}) = F(x, \mathbf{u}(x), \mathbf{v}, \mathbf{s})$$

With $\mathbf{v} \in \mathbf{R}^N, \mathbf{u} \in \mathbf{R}^M, \mathbf{s} \in \mathbf{R}^J$. Where N is the number of variables that are independent of x , M is the number of variables that depend explicitly on x and J is the number of variables that are independent of x and constant.

Let us use combined thermal conductivity K as an example. For outlined problem K is a function of form:

$$K(x, h_{comb}) = K(D_o(x), D_i(x), y(x), h_{comb}, h_{f \rightarrow w}, k_{pipe}, k_{ins})$$

If this dependence is not outlined in any form then, we state that K is constant over total length. Since, when it comes to piping, diameter does not change constantly with length, and we are dealing with segments of different length of constant diameter. We will be making a lot of reference to those discrete points, either to segment lengths or to the cumulative length up to given segment. The notation used is:

$$ith \text{ segment length: } x_i^s$$

$$\text{Cumulative length up to } nth \text{ segment: } x_n = \sum_{i=0}^n x_i^s$$

There are places where an equation can get somewhat long, and so it is helpful to introduce another notation that will help keep those equations tidy. Again, taking K as an example:

$$K_i = K(x_i, h_{comb})$$

With that out of the way, let us proceed to the fun part.

3 Derivation

3.1 Basics of steady-state heat transfer

The assumptions we must consider to make calculations possible (and easy) are:

1. Steady-state operation (both mass and energy).
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid in question is incompressible.
5. The specific heat capacity is constant.
6. No axial heat transfer.
7. Piping is horizontal and enclosed by air only at ambient pressure.

With above in mind, one can utilize "Thermal Resistance Network" methodology and write that the average heat loss rate through a pipe on a segment of length L and distance x from the inlet is:

$$\dot{Q}_{avg}(x, h_{comb}) = \frac{(T(x, h_{comb}) - T_{2\infty})_{avg}}{R(x, h_{comb}, L)} = 2\pi L K(x, h_{comb}) (T(x, h_{comb}) - T_{2\infty})_{avg} \quad (1)$$

The *avg* notation means that we should take average of the temperature difference on piping segment of length L . This is mathematically explained in equation (4). With the same assumptions we can write that energy balance at any point must satisfy:

$$\dot{m}C dT + \dot{Q} = 0 \quad (2)$$

where:

- $T(x)$ - temperature of liquid at distance x , (F°)
- $T_{2\infty}$ - ambient temperature, (F°)
- $R(L)$ - thermal resistance of piping of length L , $\left(\frac{F^\circ}{BTU}\right)$

- K - combined thermal conductivity, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$
- $\dot{m}$ - mass flow rate, $\left(\frac{lbm}{h}\right)$
- C - specific heat capacity, $\left(\frac{BTU}{lb \cdot F^\circ}\right)$

and we define K as:

$$\frac{1}{K} = 2\pi LR(L) = \frac{24}{h_{f \rightarrow w} D_i} + \frac{1}{k_{pipe}} \ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} \ln\left(\frac{D_o + 2y}{D_o}\right) + \frac{24}{h_{comb}(D_o + 2y)} \quad (3)$$

where:

- y - insulation thickness, *inch*
- D_i - inner diameter, *inch*
- D_o - outer diameter, *inch*
- $h_{f \rightarrow w}$ - heat transfer coefficient from fluid to piping wall, $\left(\frac{BTU}{ft^2 \cdot h \cdot F^\circ}\right)$
- h_{comb} - combined heat transfer coefficient from surface of a piping to air, $\left(\frac{BTU}{ft^2 \cdot h \cdot F^\circ}\right)$
- k_{pipe} - thermal conductivity of a piping, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$
- k_{ins} - thermal conductivity of a insulation, $\left(\frac{BTU}{ft \cdot h \cdot F^\circ}\right)$

3.2 General solution

For simplicity let us drop h_{comb} from functions in the derivation. So, when we write $F(x)$ we really mean $F(x, h_{comb})$. We begin our analysis with finite distance change. Using graph for this would be the simplest approach.

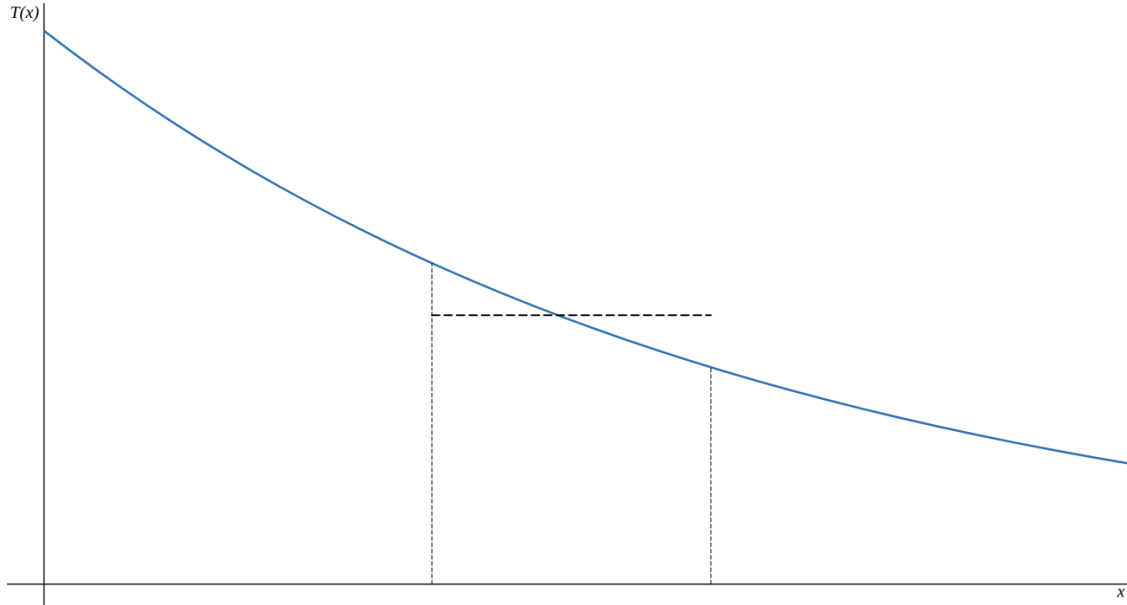


Figure 1: $T(x)$ Function of $T(x)$

In the figure above we consider a change in temperature over a finite change in length on interval $[x_i, x_{i+1}]$, where temperature drops from $T(x_i)$ to $T(x_{i+1})$. The heat loss over piping segment $x_{i+1} - x_i = \Delta x_i$ is not constant, but drops with temperature. Because of that we consider some average temperature difference to find the average heat loss. The temperature we will use to find the difference is shown on the figure above as a horizontal line between x_i and x_{i+1} . With $w_i \in [0, 1]$ we can write that the average heat loss is:

$$\dot{Q}_{avg}(x_i) = 2\pi(x_{i+1} - x_i)K_i \cdot (T(x_i) - T_{2\infty})_{avg} = 2\pi\Delta x_i K_i \cdot (T(x_i + w_i\Delta x_i) - T_{2\infty}) \quad (4)$$

And energy balance:

$$-\dot{Q}_{avg}(x_i) = \dot{m}C \cdot (T(x_{i+1}) - T(x_i)) = \dot{m}C\Delta T_i \quad (5)$$

We connect eq (4) and (5), rearrange it and write $x_i^* = x_i + w_i\Delta x_i$:

$$\frac{1}{T(x_i^*) - T_{2\infty}} \frac{\Delta T_i}{\Delta x_i} = -\frac{2\pi K_i}{\dot{m}C} = -\frac{K_i}{\lambda} \quad (6)$$

Remembering that by $T(x)$ we mean $T(x, h)$ we can finally write:

$$\lim_{\max(\Delta x_i) \rightarrow 0} \frac{1}{T(x_i^*, h) - T_{2\infty}} \left(\frac{\Delta T_i}{\Delta x_i} \right)_h = \frac{1}{T(x, h) - T_{2\infty}} \frac{\partial T(x, h)}{\partial x} = -\frac{K(x, h)}{\lambda} \quad (7)$$

And the general solution after some cleaning can be written as:

$$\frac{\partial \ln(T(x, h) - T_{2\infty})}{\partial x} = -\frac{K(x, h)}{\lambda} \quad (8)$$

Integrating LHS from T_0 to $T(x, h)$ and RHS from 0 to x yields:

$$\ln \left(\frac{T(x, h) - T_{2\infty}}{T_0 - T_{2\infty}} \right) = -\frac{1}{\lambda} \int_0^x K(l, h) dl \quad (9)$$

To solve it we need to know how $K(x, h)$ changes with length. We would also need to show that h is not a function of x . If we take a look at K it's obvious we can vary h independently of x , but we assumed steady-state conditions, and if this forces h to be dependent on x then it would render solving this integral hard or even impossible.

I was not able to prove it mathematically but in every tested scenario I tried h is independent of x . All my calculation and methodology used is described in chapter (X). For the time being we assume that h is independent and proceed with solving eq (9) for each case.

ANOTHER TAKE: I was not able to prove it mathematically but I based my argument on statistical interpretation of our assumptions as well as results of used heat transfer coefficients correlations. All my calculation and methodology used is described in chapter (X). For the time being we assume that h is independent and proceed with solving eq (9) for each case.

3.3 Solutions $h_{comb} \rightarrow \infty$

We start our approach to solving eq (9) by letting $h_{comb} \rightarrow \infty$. This eliminates h_{comb} from K providing us with simple solutions. The main consequence is that surface temperature of insulation is driven to ambient temperature around piping, which maximizes temperature gradient maximizing heat transfer in turn. This method provides the most conservative solution and yields the biggest temperature drop. This method was used in my excel template for the company I'm working for as "conservative" approach, providing user with quick and simple answer to the question "is currently selected insulation thickness enough for the worst-case scenario?".

3.3.1 Constant diameter

We start with a long pipe of length L_t of constant inner and outer diameter D_i, D_o , respectively as well as constant insulation thickness y . Those assumptions allows us to say that K does not depend on x and so we can take it out of the integral. The solution is, then:

$$\ln \left(\frac{T(x) - T_{2\infty}}{T_0 - T_{2\infty}} \right) = -\frac{K}{\lambda} x \quad (10)$$

for $x \in [0, L_t]$. Short and sweet. Not much to say, and so we move to the next scenario.

3.3.2 Segments with different diameter and insulation thickness

Here we let D_i, D_o and y vary over specified segments with length. We have to figure out the form of $K(x)$. It is of course just a piecewise defined function:

$$K(x) = \begin{cases} K(x_0) & \text{if } 0 < x \leq x_0 \\ \cdot & \\ \cdot & \\ \cdot & \\ K(x_n) & \text{if } x_{n-1} < x \leq x_n \end{cases} \quad (11)$$

This form provides a simple way of solving the integral from eq (9), but writting a python code around it would be challenging and convoluted, at least that was my fear. Because of that I went with redefinind $K(x)$ with help of modified heaviside function:

$$H(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (12)$$

Here, I have to confess that I struggled a lot with the choice of heaviside funciton 0th value. Intuitively, it shouldn't matter if at $H(0) = 0$ or $H(0) = 1$, but I was getting different results. I thus, analysed how does AFT Fathom calculates points of discontinuity and found out the solution. It is described in more detail in chapter (X).

This choice of heaviside function introduces point of discontinuity in every property. This is of course not the case in reality, as the change is done with reducers provoding continuous change in diameter as well as properties. We could of course have 90 degrees reducers and one wouldn't expect discontinuity there as well, and this is true, this would be the place where axial heat transfer would be present, and we would observe that beginning of smaller diameter pipe is hotter than the average of the whole segment.

Intrestingly enough we could use different connecting function instead of heaviside, like logistic funtion or clamping function between 0 and 1. But the integration and python implementation is not so straightforward as with heaviside function. With that we can define $K(x)$ as:

$$K(x) = \sum_{i=1}^n (K_i - K_{i-1}) H(x - x_{i-1}) \quad (13)$$

Nice thing about above equation is that it is easy to integrate, simple to vectorize with right shifting function and writing it in python is straightforward. And so we find that the solution of eq (9) is:

$$\begin{aligned}
\ln \left(\frac{T(x) - T_{2\infty}}{T_0 - T_{2\infty}} \right) &= -\frac{1}{\lambda} \int_0^x K(l) dl \\
&= -\frac{1}{\lambda} \int_0^x \sum_{i=1}^n (K_i - K_{i-1}) H(l - x_{i-1}) dl \\
&= -\frac{1}{\lambda} \sum_{i=1}^n (x - x_{i-1}) (K_i - K_{i-1}) H(x - x_{i-1}) \\
&= -\frac{1}{\lambda} (x\mathbf{1} - s(\mathbf{x})) \odot (\mathbf{K} - s(\mathbf{K})) \cdot H(x\mathbf{1} - s(\mathbf{x}))
\end{aligned} \tag{14}$$

where $s(\cdot)$ is a right shifting function that has a property $s(\mathbf{x}) = s([x_i]_{i=0}^n) = [0, x_i]_{i=0}^{n-1}$ and $\mathbf{1}$ is a column vector of ones. We can interchange integral with summation, because the function is constant on each segment interval $[x_{i-1}^c, x_i^c]$.

This is the "conservative approach" method, yielding the biggest temperature drop accross the whole pipe. It is worth mentioning that for thick enough insulation, this method gives preety accurate results, but "thick enough insulation" is enought of an ambiguous statement to not mean anything in particular. When thickness of insulation is close to 0, this method brings fulid temperature to ambient temperature in matters of meters, and would be highly disregarded for those type of applications where accuracy is needed.

3.4 Exact solutions